

MA1513 Exam (Exemplify)

AY2024/2025 Semester II

MCQ/FIB (15 questions; 3 marks each)

1. Let $\mathbf{Ax} = \mathbf{b}$ be a non-homogeneous linear system where $\mathbf{R}$ is the reduced row echelon form (inclusive of the augmented column) of the augmented matrix $(\mathbf{A} \mid \mathbf{b})$. Which of the following statements must be true? (Pick all correct options)
- i. If every column of $\mathbf{R}$ is a pivot column, then $\mathbf{Ax} = \mathbf{b}$ is inconsistent.
 - ii. If $\mathbf{R}$ has a non-pivot column, then $\mathbf{Ax} = \mathbf{b}$ has infinitely many solutions.
 - iii. If $\mathbf{R}$ has a zero row, then $\mathbf{Ax} = \mathbf{b}$ is consistent.
 - iv. If every row of $\mathbf{R}$ has a leading entry, then $\mathbf{Ax} = \mathbf{b}$ has exactly one solution.
 - v. If $\mathbf{R}$ does not have a leading entry in the last column, then $\mathbf{Ax} = \mathbf{b}$ is consistent.

Explanation:

- (i) As long as the augmented column of $\mathbf{R}$ (corresponding to the constant terms) is a pivot column, the system must be inconsistent.
- (ii) If the only non-pivot column is the augmented column, the system will have exactly one solution.
- (iii) It is possible for $\mathbf{R}$ to have a row $(0 \ 0 \ \dots \ 0 \mid 1)$ above the zero row.
- (iv) If the row has the leading entry in the augmented column, it will look like $(0 \ 0 \ \dots \ 0 \mid 1)$.
- (v) This means the last column is a non-pivot column, and hence the system is consistent.

2. Which of the following are equivalent conditions for an $n \times n$ matrix $\mathbf{A}$ to be non-singular? (Pick all correct options)
- i. The n rows of $\mathbf{A}$ are linearly independent.
 - ii. Any row echelon form of $\mathbf{A}$ has n pivot columns.
 - iii. The column space of $\mathbf{A}$ is $\mathbb{R}^n$.
 - iv. The determinant of $\mathbf{A}$ is 1.
 - v. The non-homogeneous system $\mathbf{Ax} = \mathbf{b}$ has only the trivial solution.

Explanation:

- (i) If the n rows of $\mathbf{A}$ are linearly independent, then all the rows in the RREF $\mathbf{R}$ must be nonzero, and hence $\mathbf{R}$ must be the identity matrix. So $\mathbf{A}$ is non-singular.
- (ii) If REF of $\mathbf{A}$ has n pivot columns, then $\mathbf{Ax} = \mathbf{0}$ will only have trivial solution. So $\mathbf{A}$ is non-singular.
- (iii) If the column space of $\mathbf{A}$ is $\mathbb{R}^n$, all the n columns of $\mathbf{A}$ form a basis for the column space. This means all the n columns of REF of $\mathbf{A}$ are pivot columns. By (ii), $\mathbf{A}$ is non-singular.
- (iv) While determinant is 1 guaranteed $\mathbf{A}$ is non-singular, the converse need not be true. i.e. knowing $\mathbf{A}$ is non-singular, does not guarantee $\det(\mathbf{A}) = 1$. Hence it is not an “equivalent” condition.
- (v) The statement is flawed. A non-homogeneous system cannot have trivial solution.

3. Given matrix $\mathbf{A} = \begin{pmatrix} a & b & c \\ i & j & k \\ x & y & z \end{pmatrix}$ has a non-zero determinant n . Which of the following are correct? (Pick all correct options)

i. $\begin{vmatrix} a & b & c \\ 2i & 2j & 2k \\ x & y & z \end{vmatrix} = 2^3 n$

$$\begin{aligned}
 \text{ii. } & \begin{vmatrix} x & y & z \\ a & b & c \\ i & j & k \end{vmatrix} = -n \\
 \text{iii. } & \begin{vmatrix} z & y & x \\ k & j & i \\ c & b & a \end{vmatrix} = n \\
 \text{iv. } & \begin{vmatrix} a & 3i & 5x \\ b & 3j & 5y \\ c & 3k & 5z \end{vmatrix} = 15n \\
 \text{v. } & \begin{vmatrix} a-i & b-j & c-k \\ i-x & j-y & k-z \\ x-a & y-b & z-c \end{vmatrix} = n
 \end{aligned}$$

Explanation:

While it is possible to expand the 3x3 determinants in each option to check the answers, there are some shortcuts to do so.

(i) We can take out the “common factor” 2 from the second row of the determinant. So the correct answer should be 2n.

(ii) This determinant is obtained from A by doing two row interchanging operations: R1 & R2, followed by R1 & R3. Each row interchange contributes a -1 factor to the determinant. So the resulting determinant should be $(-1) \times (-1) \times n = n$.

(iii) This determinant is obtained from A by doing one row interchange (R1 & R3), and one column interchange (C1 & C3), each contributing one -1 factor to the determinant. So the resulting determinant is $(-1) \times (-1) \times n = n$.

(iv) We can take out the “common factor” 3 from the second column, and “common factor” 5 from the third column of the determinant. So the correct answer is $3 \times 5 \times n = 15n$.

(v) Observe that the third row of the determinant is the negative of the sum of the first and second row. This means the three rows are linearly dependent, and hence the determinant should be 0.

4. A is an $m \times n$ matrix which is full rank. Which of the following are correct? (Pick all correct options.)

- If $m < n$, it is always possible to add an additional row to **A** so that the resulting matrix has a larger rank.
- If $m < n$, it is always possible to add an additional column to **A** so that the resulting matrix has a larger rank.
- If $m > n$, it is always possible to add an additional row to **A** so that the resulting matrix has a larger rank.
- If $m > n$, it is always possible to add an additional column to **A** so that the resulting matrix has a larger rank.
- If $m = n$, adding an additional row or column to **A** will result in a matrix with a larger rank.

Explanation:

(i) $m < n$ means A is rank m (since it is full rank). So there are columns without leading entries, and hence we can still add rows with leading entries at those “missing” positions to increase the rank.

- (ii) Since all the rows in the REF of A has a leading entry, by adding columns will not increase the number of leading entries, and hence no change to the rank.
- (iii) Since all the columns in the REF of A has a leading entry, by adding rows will not increase the number of leading entries, and hence no change to the rank.
- (iv) $n < m$ means A is rank n (since it is full rank). So there are rows without leading entries, and hence we can still add columns with leading entries at those “missing” positions to increase the rank.
- (v) $n = m$ means A is rank n ($= m$). Since all the rows/columns in the REF of A has a leading entry, by adding rows/columns will not increase the number of leading entries, and hence no change to the rank.

5. Which of the following represent the plane $x - y + 2z = 0$ in the xyz-space?

(Pick all correct options)

- i. $\text{span}\{ (1, -1, 2), (0, 0, 0) \}$
- ii. $\text{span}\{ (1, 1, 0), (0, 2, 1), (2, 0, -1) \}$
- iii. $\text{span}\{ (1, 3, 1), (3, 9, 3) \}$
- iv. $\text{span}\{ (3, 1, -1), (3, -1, -2) \}$
- v. $\text{span}\{ (1, 0, 0), (0, -1, 0), (0, 0, 2) \}$

Explanation:

We check all the vectors in each linear span lie on the plane (by substituting the vectors into the equation and see if they all satisfy the equation. (Vectors in (i) and (v) are not.) We should also make sure there are two vectors in each set that are linearly independent to each other. (Vectors in (iii) are not.)

6. Which of the following sets are linearly independent? (Pick all correct options)

- i. $\{ (10, 11, 12), (100, 121, 144) \}$
- ii. $\{ (1, 5, 1, 3), (1, 3, 1, 5), (5, 1, 3, 1), (3, 1, 5, 1) \}$
- iii. $\{ (3, 6, 9), (13, 16, 19), (23, 26, 29) \}$
- iv. $\{ (0, 1, 2, 3), (0, 0, 1, 2), (0, 0, 0, 3) \}$
- v. $\{ (1, 0, 1, 0), (0, 2, 0, 2), (3, 3, 3, 3) \}$

Explanation:

- (i) The two vectors are not scalar multiple of each other
- (ii) The 4 vectors stack to form a 4×4 matrix with nonzero determinant.
- (iii) Observe that $v_3 - v_1 = 2(v_2 - v_1)$. So there is a linear relationship among the three vectors.
- (iv) Check using standard approach that the vector equation has only trivial solution.
- (v) Observe that $3v_1 + (1/2)v_2 = v_3$. So there is a linear relationship among the three vectors.

7. A 3×5 matrix C has columns c_1, c_2, c_3, c_4, c_5 arranged from left to right. Suppose the reduced

row echelon form of C is $\begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. Which of the following sets are bases for the column space of C ? (Pick all correct options)

- i. $\{ c_1, c_3, c_5 \}$
- ii. $\{ c_1, c_2, c_5 \}$
- iii. $\{ c_1, c_3, c_4 \}$
- iv. $\{ c_3, c_4, c_5 \}$

- v. Any set of 3 linearly independent vectors in $\mathbf{R}^3$.

Explanation:

Since there are three pivot columns in the RREF of C, the dimension of the column space is 3. This implies the column space is equal to $\mathbf{R}^3$. So any three linearly independent vectors form a basis for the column space.

(i) Since columns 1, 3, 5 of the RREF are pivot columns, this is a basis.

(ii) Since columns 1 and 2 of the RREF are the same, $\mathbf{c}_1$ and $\mathbf{c}_2$ of matrix C are also the same.

(iii) Since column 4 of the RREF is the sum of columns 1 and 3, we also have $\mathbf{c}_4 = \mathbf{c}_1 + \mathbf{c}_3$ in matrix C.

(iv) Since columns 3, 4 and 5 of the RREF are linearly independent, the corresponding $\mathbf{c}_3, \mathbf{c}_4, \mathbf{c}_5$ are also linearly independent.

(v) Since the column space of C = $\mathbf{R}^3$, any 3 linearly independent vectors is a basis.

8. Given $S = \{\mathbf{u}_1, \mathbf{u}_2\}$ is a basis for $\mathbf{R}^2$. Suppose the coordinate vector of $\mathbf{v} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$ with respect to S is given by $(\mathbf{v})_S = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and the coordinate vector of $\mathbf{w} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$ with respect to S is given by $(\mathbf{w})_S = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Then $\mathbf{u}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{u}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ where $a = \underline{\hspace{1cm}}, b = \underline{\hspace{1cm}}, c = \underline{\hspace{1cm}}, d = \underline{\hspace{1cm}}$ (FIB)

Explanation:

We have $2\begin{pmatrix} a \\ b \end{pmatrix} - \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$ and $-\begin{pmatrix} a \\ b \end{pmatrix} + 2\begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$.

Solve for a, b, c and d.

9. Let $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be a linear transformation such that all the vectors along the line $y = 2x$ is scaled by a factor 2, and all the vectors along the line $y = -2x$ is scaled by a factor -2. Then the standard matrix of T is given by $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where $a = \underline{\hspace{1cm}}, b = \underline{\hspace{1cm}}, c = \underline{\hspace{1cm}}, d = \underline{\hspace{1cm}}$ (FIB)

Explanation:

A point on $y = 2x$ is $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. We have $T\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$.

A point on $y = -2x$ is $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$. We have $T\left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}\right) = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$.

By stacking method, $A\begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 4 & 4 \end{pmatrix}$ which gives $A = \begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}$.

10. Let $\mathbf{A}$ be a 4 x 4 matrix. Suppose two of its eigenvalues are 3 and 0 with corresponding

$$\text{eigenspaces } E_3 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}\right\} \text{ and } E_0 = \text{span}\left\{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\right\}.$$

Which of the following statements must be true? (Pick all correct options)

- $\mathbf{A}$ has exactly 2 eigenvalues.
- The multiplicity of the eigenvalue 0 is equal to 1.
- $\mathbf{A}$ may be a non-diagonalizable matrix.
- Dimension of E_3 is equal to 3.
- $\lambda = 3$ is a repeated root of the characteristic polynomial of $\mathbf{A}$.

Explanation:

Note that the three vectors in the span of E_3 are linearly dependent (the third vector is a linear combination of the first two).

(i) It is possible that there is a third eigenvalue other than 3 and 0.

(ii) It is possible that the multiplicity of eigenvalue 0 is equal to 2.

(iii) If 3 and 0 are the only two eigenvalues of A , then there are only 3 linearly independent eigenvectors for A . In this case, A will be non-diagonalizable.

(iv) After removing the redundant vector in the span, we see that the dimension of E_3 is 2.

(v) Since the dimension of E_3 is 2, the multiplicity of eigenvalue 3 must be at least 2, and hence it is a repeated root.

11. Suppose $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ where $\mathbf{P} = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}$ where a and b are distinct.

Which of the following are true? (Pick all correct options)

- i. $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} A \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} b & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix}$
- ii. $\begin{pmatrix} 1 & 0 & 4 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{pmatrix}^{-1} A \begin{pmatrix} 1 & 0 & 4 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & 2b \end{pmatrix}$
- iii. $\begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}^{-1} A \begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}$
- iv. $\begin{pmatrix} 1 & 2 & 4 \\ 1 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}^{-1} A \begin{pmatrix} 1 & 2 & 4 \\ 1 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & b \end{pmatrix}$
- v. $\begin{pmatrix} 2 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}^{-1} A \begin{pmatrix} 2 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & a \end{pmatrix}$

Explanation:

a is an eigenvalue with eigenspace $\text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$ and b is an eigenvalue with eigenspace $\text{span}\left\{\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}\right\}$.

(i) The three columns of $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are the same as $\mathbf{P}$ with rearrangement, so that the diagonal entries of the resulting diagonal matrix is b, a, a in that order.

(ii) For $\begin{pmatrix} 1 & 0 & 4 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{pmatrix}$, the first two columns are also eigenvectors with eigenvalue a , and the third column is still the eigenvector with eigenvalue b . So the diagonal entries of the resulting diagonal matrix is a, a, b in that order.

(iii) The third column of $\begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ is not an eigenvector. So the matrix does not diagonalize A .

(iv) The second and third columns of $\begin{pmatrix} 1 & 2 & 4 \\ 1 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}$ are linearly dependent. Hence the matrix is not a non-singular matrix.

(v) The first and third columns of $\begin{pmatrix} 2 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$ are eigenvectors with eigenvalue a , and the second column is an eigenvector with eigenvalue b . So the diagonal entries of the resulting diagonal matrix is a, b, a in that order.

12. Given a subspace $V = \text{span}\{(1, 1, 0, 0), (0, 0, 1, 1)\}$ and a vector $\mathbf{u} = (3, -1, 2, -4)$ in $\mathbb{R}^4$. The vector $\mathbf{v}$ in V that is closest to $\mathbf{u}$ is given by $\mathbf{v} = (a, b, c, d)$ where $a = \underline{-1}, b = \underline{1}, c = \underline{-1}, d = \underline{-1}$ (FIB)

Explanation:

The vector $\mathbf{v}$ is the projection of vector $\mathbf{u}$ onto the column space of $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$.

First find the least squares solution of $\mathbf{Ax} = \mathbf{u}$, which gives $\mathbf{x}_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Then $\mathbf{v} = \mathbf{Ax}_0 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$.

13. Given a 2×2 matrix $\mathbf{A}$ with only 1 eigenvalue $\lambda = -1$ and only 1 linearly independent eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Suppose the linear system $(\mathbf{A} + \mathbf{I})\mathbf{u} = \mathbf{v}$ has a general solution $\mathbf{u} = \begin{pmatrix} 1 + s \\ s \end{pmatrix}$. Which of the following are solutions of the SDE $\mathbf{y}' = \mathbf{Ay}$?

i. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

ii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

iii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-t} \begin{pmatrix} t \\ -t - 1 \end{pmatrix}$

iv. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-t} \begin{pmatrix} 1 + t \\ t \end{pmatrix}$

v. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-t} \begin{pmatrix} t + 2 \\ -t - 1 \end{pmatrix}$

Explanation:

(i) $\mathbf{w} = -\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$. So $\mathbf{w}e^{\lambda t}$ is a solution of the SDE in the standard form.

(ii) $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (with $s = 0$) is a generalised eigenvector of $\mathbf{A}$ with respect to $\mathbf{v}$. But $\mathbf{u}e^{\lambda t}$ does not give a solution of the SDE.

(iii) $\mathbf{u} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ (with $s = -1$) is a generalised eigenvector of $\mathbf{A}$ with respect to $\mathbf{v}$. So $(\mathbf{vt} + \mathbf{u})e^{\lambda t}$ is a solution of the SDE.

(iv) $\mathbf{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is not an eigenvector of $\mathbf{A}$ while $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (with $s = 0$) is a generalised eigenvector of $\mathbf{A}$ with respect to $\mathbf{v}$. So $(\mathbf{w}t + \mathbf{u})e^{\lambda t}$ is not a solution of the SDE.

(v) $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$ and $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is a generalised eigenvector of $\mathbf{A}$ with respect to $\mathbf{v}$. So $\mathbf{v}e^{\lambda t}$ and $(\mathbf{v}t + \mathbf{u})e^{\lambda t}$ are solutions of the SDE, and the sum $\mathbf{v}e^{\lambda t} + (\mathbf{v}t + \mathbf{u})e^{\lambda t}$ is also a solution of the SDE.

14. Given an SDE $\mathbf{y}' = \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix} \mathbf{y}$. Which of the following are solutions of the SDE?

- i. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \frac{1}{2} \cos t + \frac{1}{2} \sin t \\ \cos t \end{pmatrix}$
- ii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \frac{1}{2} \cos t - \frac{1}{2} \sin t \\ \cos t \end{pmatrix}$
- iii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} -\sin t + \cos t \\ -2 \sin t \end{pmatrix}$
- iv. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \sin t + \cos t \\ 2 \sin t \end{pmatrix}$
- v. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \frac{1}{2} \sin t - \frac{1}{2} \cos t \\ -\sin t \end{pmatrix}$

Explanation:

The eigenvalues of the underlying matrix is $r_1 = 1 - i$ and $r_2 = 1 + i$ with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} \frac{1}{2}(1 + i) \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} \frac{1}{2}(1 - i) \\ 1 \end{pmatrix}$.

Take $a = 1$, $b = -1$, $\mathbf{p} = \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$.

(i) This has the form $e^{at}[\cos bt \mathbf{p} - \sin bt \mathbf{q}]$, which is a real solution of the SDE.

(ii) This has the form $e^{at}[\cos bt \mathbf{p} + \sin bt \mathbf{q}]$, which is not a solution of the SDE.

(iii) This has the form $2e^{at}[\sin bt \mathbf{p} + \cos bt \mathbf{q}]$, which is a real solution of the SDE.

(iv) This has the form $2e^{at}[-\sin bt \mathbf{p} + \cos bt \mathbf{q}]$, which is not a solution of the SDE.

(v) This has the form $e^{at}[\sin bt \mathbf{p}' + \cos bt \mathbf{q}']$, where $\mathbf{p}' = \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix}$ and $\mathbf{q}' = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}$ which is not a solution of the SDE.

15. Suppose $\mathbf{A}$ is a 2×2 non-zero matrix which is singular and diagonalizable. Which of the following are correct about the SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$? (Pick all correct options)

- i. The equilibrium point can be a centre.
- ii. The equilibrium point cannot be a degenerate node.
- iii. The equilibrium point can be a star node.
- iv. There are trajectories in the phase portrait that are straight lines.
- v. There are trajectories in the phase portrait that are not straight lines.

Explanation:

Since $\mathbf{A}$ is singular, it has an eigenvalue 0, so its equilibrium point is the exceptional case. So it cannot be centre or star node. Since the matrix is non-zero and diagonalizable, the other eigenvalue is non-zero. So the equilibrium point cannot be degenerate, and the phase portrait only consists of straight-line trajectories.

Student number (9 digits starting with A)

Assigned Seat number (3 digits)

NATIONAL UNIVERSITY OF SINGAPORE

MA1513 Linear Algebra with Differential Equations

FINAL EXAM

Semester 2 AY2024/25

Time Allowed: 1.5 Hours

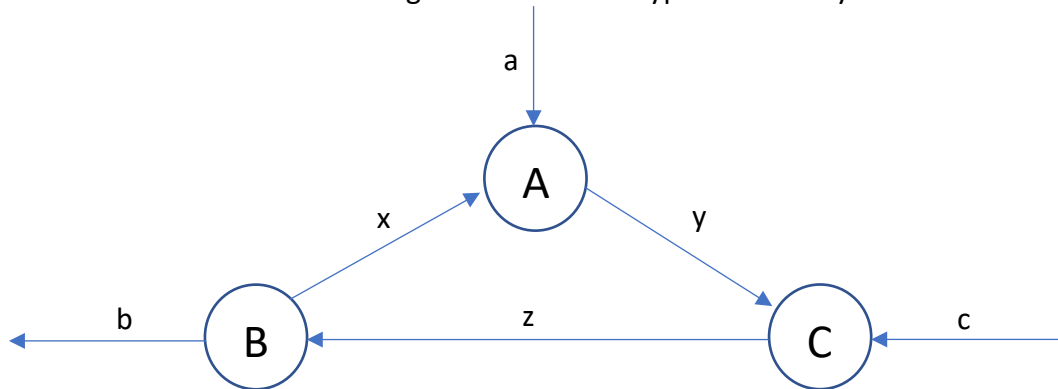
INSTRUCTIONS TO STUDENTS

1. Write down your student number and assigned seat number at the top of this page.
DO NOT WRITE YOUR NAME!
2. This paper contains **FOUR** written questions and comprises **FIFTEEN** printed pages.
3. Students are required to answer the MCQ questions (Q1 to Q15) directly in Exemplify, and write the answers with workings of the written questions (Q16 to Q19) in this booklet.
4. This is an OPEN BOOK exam.
5. Students are allowed to use MATLAB that are installed in their PC.
6. Students are allowed to use any calculators.

Instructions: You should write your working and answers in the spaces provided in this exam booklet only. Write the final answers in the specific boxes provided. You may use MATLAB for all the questions. But you should show how your answers are derived.

Question 16 (10 marks)

The network in the diagram shows the traffic flow (in vehicles per hour) over several one-way streets in the downtown area during rush hours on a typical work day.



- (i) (3 marks) Suppose the traffic volumes for a , b , c are 100, 150 and 50 respectively. Set up a linear system $\mathbf{Ax} = \mathbf{b}$ with 3 equations (in the order of the junctions A, B and C) with three variables x , y , z arranged in that order. Write down the matrix $\mathbf{A}$ and vector $\mathbf{b}$ explicitly. All entries of vector $\mathbf{b}$ should be positive.

Working space

$$\text{A: } a + x = y \rightarrow -x + y = a = 100$$

$$\text{B: } z = b + x \rightarrow -x + z = b = 150$$

$$\text{C: } c + y = z \rightarrow -y + z = c = 50$$

$$\begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 100 \\ 150 \\ 50 \end{pmatrix}$$

Write the matrices $\mathbf{A}$ and $\mathbf{b}$ in the box

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 100 \\ 150 \\ 50 \end{pmatrix}$$

(ii) (2 marks) Find the general solution of the linear system in (i). (You may use MATLAB for the working.)

Working space

$$\left(\begin{array}{ccc|c} -1 & 1 & 0 & 100 \\ -1 & 0 & 1 & 150 \\ 0 & -1 & 1 & 50 \end{array}\right) \xrightarrow{GJE} \left(\begin{array}{ccc|c} 1 & 0 & -1 & -150 \\ 0 & 1 & -1 & -50 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Let $z = t$. Then $y = t - 50$ and $x = t - 150$.

Write the general solution in the box

$$x = t - 150$$

$$y = t - 50$$

$$z = t$$

(iii) (2 marks) Given the traffic volumes in (i), what is the smallest possible value of y ?

Working space

Since x, y, z must all be non-negative,

we need to have $t \geq 150$.

So the smallest possible value of $y = 150 - 50 = 100$.

Write only the number in the box

100

(iv) (3 marks) If we are not given the specific traffic volumes for a , b and c in (i), what is the condition (in terms of an equation in a , b and c) so that there are solutions for x , y , z ?

Working space

From the augmented matrix $\left(\begin{array}{ccc|c} -1 & 1 & 0 & a \\ -1 & 0 & 1 & b \\ 0 & -1 & 1 & c \end{array}\right)$, by Gaussian elimination we have

$$\left(\begin{array}{ccc|c} -1 & 1 & 0 & a \\ -1 & 0 & 1 & b \\ 0 & -1 & 1 & c \end{array}\right) \xrightarrow{R_2 - R_1} \left(\begin{array}{ccc|c} -1 & 1 & 0 & a \\ 0 & -1 & 1 & b - a \\ 0 & -1 & 1 & c \end{array}\right) \xrightarrow{R_3 - R_2} \left(\begin{array}{ccc|c} -1 & 1 & 0 & a \\ 0 & -1 & 1 & b - a \\ 0 & 0 & 0 & c - b + a \end{array}\right)$$

For the system to have solutions, we need $c - b + a = 0$.

Write the equation in the box

$$c - b + a = 0$$

Additional working space for Q16 (Indicate the parts clearly)

Question 17 (10 marks)

A damped mass-spring system is given by a 2nd order ODE: $my'' + cy' + ky = 0$, where k is the spring constant (in Newton per meter N/m) and c is the damping constant, while m is the mass (in kg) of the ball. Three sets of readings for displacement y in meter (m), velocity y' in m/s and acceleration y'' in m/s^2 are measured and recorded as follow:

Measurements	y (m) *	y' (m/s)	y'' (m/s ²)
1 st set	0 (equilibrium position)	1.5 (downward)	-12.5
2 nd set	0.5 (below e. p.)	0.5 (downward)	-15
3 rd set	0.2 (below e. p.)	-1 (upward)	10

* Value for y is positive below the e. p., and negative above the e. p.

(i) (4 marks) Suppose the mass of the ball is 1kg. Use the above measurements to set up a linear system $\mathbf{Ax} = \mathbf{b}$ to solve for c and k .

Working space

Substitute the measurements to the ODE

(with $m = 1$):

$$-12.5 + 1.5c + 0k = 0 \rightarrow 1.5c + 0k = 12.5$$

$$-15 + 0.5c + 0.5k = 0 \rightarrow 0.5c + 0.5k = 15$$

$$10 - c + 0.2k = 0 \rightarrow -c + 0.2k = -10$$

Write the matrices $\mathbf{A}$ and $\mathbf{b}$ in the box

$$\mathbf{A} = \begin{pmatrix} 1.5 & 0 \\ 0.5 & 0.5 \\ -1 & 0.2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 12.5 \\ 15 \\ -10 \end{pmatrix}$$

(ii) (2 marks) Write down the ODE that best fit the above readings.

Working space

Find the least squares solution of $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} c \\ k \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} (\mathbf{A}^T \mathbf{b}) = \begin{pmatrix} 10.11 \\ 17.22 \end{pmatrix}$$

Write the best fit ODE in the box
(give your c and k up to 2 decimal places)

$$y'' + 10.11 y' + 17.22 y = 0$$

(iii) (2 marks) Is the system over-damping or under-damping? Why?

Working space

$$m = 1, c = 10.11 \text{ and } k = 17.22$$

$$c^2 = 102.21$$

$$4km = 68.88$$

Write the answer and explanation in the box

Since $c^2 > 4km$, the system is over-damping.

(iv) (2 marks) Convert the ODE into an SDE without solving it. (You are not required to show the working.)

Write the SDE in the box (give your answer up to 2 decimal places)

$$y' = \begin{pmatrix} 0 & 1 \\ -17.22 & -10.11 \end{pmatrix} y$$

Additional working space for Q17 (Indicate the parts clearly.)

Question 18 (12 marks)

Three coffee brands A, B and C are available to the 30,000 households in city X. The monthly market

shares (in terms of number of households) of the three brands is modelled by $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$ with $\mathbf{x}_n = \begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix}$

where a_n , b_n and c_n are the number of households purchasing brands A, B and C respectively in the n^{th} month. (Assuming each household only purchase one coffee brand per month.)

According to the market research, the market shares in January to April 2024 are as follow:

Brand	Jan	Feb	Mar	Apr
A	10000	11200	11740	11968
B	8000	9800	10760	11282
C	12000	9000	7500	6750

(i) (4 marks) Find the stage matrix **A**.

Working space

$$A \begin{pmatrix} 10000 \\ 8000 \\ 12000 \end{pmatrix} = \begin{pmatrix} 11200 \\ 9800 \\ 9000 \end{pmatrix}$$

$$A \begin{pmatrix} 11200 \\ 9800 \\ 9000 \end{pmatrix} = \begin{pmatrix} 11740 \\ 10760 \\ 7500 \end{pmatrix}$$

$$A \begin{pmatrix} 11740 \\ 10760 \\ 7500 \end{pmatrix} = \begin{pmatrix} 11968 \\ 11282 \\ 6750 \end{pmatrix}$$

By stacking,

$$A \begin{pmatrix} 10000 & 11200 & 11740 \\ 8000 & 9800 & 10760 \\ 12000 & 9000 & 7500 \end{pmatrix} = \begin{pmatrix} 11200 & 11740 & 11968 \\ 9800 & 10760 & 11282 \\ 9000 & 7500 & 6750 \end{pmatrix}$$

$$A = \begin{pmatrix} 11200 & 11740 & 11968 \\ 9800 & 10760 & 11282 \\ 9000 & 7500 & 6750 \end{pmatrix} \begin{pmatrix} 10000 & 11200 & 11740 \\ 8000 & 9800 & 10760 \\ 12000 & 9000 & 7500 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 0.8 & 0.1 & 0.2 \\ 0.1 & 0.8 & 0.2 \\ 0.1 & 0.1 & 0.6 \end{pmatrix}$$

Write the matrix **A** in the box

$$\mathbf{A} = \begin{pmatrix} 0.8 & 0.1 & 0.2 \\ 0.1 & 0.8 & 0.2 \\ 0.1 & 0.1 & 0.6 \end{pmatrix}$$

(ii) (3 marks) Use MATLAB command to find a matrix **P** that diagonalizes **A**, together with the diagonal matrix **D**. (You are not required to show workings.)

Write the matrices **P** and **D** in the box (give your answers in fractions)

$$\mathbf{P} = \begin{pmatrix} -1 & 2 & -1/2 \\ 1 & 2 & -1/2 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\mathbf{D} = \begin{pmatrix} 7/10 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}$$

(iii) (3 marks) Use diagonalization to find the market shares of the three brands in the long run. Show your workings.

Working space

$$A^n = P D^n P^{-1} x_0$$

$$= \begin{pmatrix} -1 & 2 & -1/2 \\ 1 & 2 & -1/2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} (7/10)^n & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (1/2)^n \end{pmatrix} \begin{pmatrix} -1 & 2 & -1/2 \\ 1 & 2 & -1/2 \\ 0 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 10000 \\ 8000 \\ 12000 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 2 & -1/2 \\ 1 & 2 & -1/2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 2 & -1/2 \\ 1 & 2 & -1/2 \\ 0 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 10000 \\ 8000 \\ 12000 \end{pmatrix}$$

$$= \begin{pmatrix} 2/5 & 2/5 & 2/5 \\ 2/5 & 2/5 & 2/5 \\ 1/5 & 1/5 & 1/5 \end{pmatrix} \begin{pmatrix} 10000 \\ 8000 \\ 12000 \end{pmatrix} = \begin{pmatrix} 12000 \\ 12000 \\ 6000 \end{pmatrix}$$

Write the answer in the box

$$a_n = 12000$$

$$b_n = 12000$$

$$c_n = 6000$$

(iv) (2 marks) True or false: Regardless of the initial market shares a_0, b_0 and c_0 , it is not possible for the three coffee brands to have equal market shares in the long term.

Working space

$$\begin{pmatrix} 2/5 & 2/5 & 2/5 \\ 2/5 & 2/5 & 2/5 \\ 1/5 & 1/5 & 1/5 \end{pmatrix} \begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix} = \begin{pmatrix} \frac{2}{5}(a_0 + b_0 + c_0) \\ \frac{2}{5}(a_0 + b_0 + c_0) \\ \frac{1}{5}(a_0 + b_0 + c_0) \end{pmatrix}$$

Write either True or False in the box

True

Additional working space for Q18 (Indicate the parts clearly.)

Question 19 (13 marks)

(a) Given SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -5 & 4 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ where the underlying matrix has two linearly independent eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

(i) (3 marks) Write down a fundamental set of solutions of the SDE using the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

Working space

$$\begin{pmatrix} -5 & 4 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} -5 & 4 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = -3 \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

So eigenvalues of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ are 3 and -3 respectively.

Write the answer in the box

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} \quad \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-3t}$$

(ii) (3 marks) Solve the initial value problem with the above SDE and the initial values $y_1(0) = 3$ and $y_2(0) = 0$.

Working space

General solution of SDE:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-3t}$$

Substituting the initial value:

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

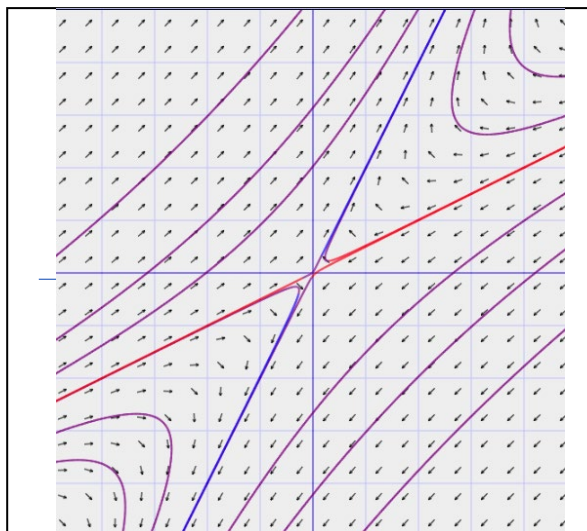
This gives $c_1 = -1$ and $c_2 = 2$.

Write the answer in the box

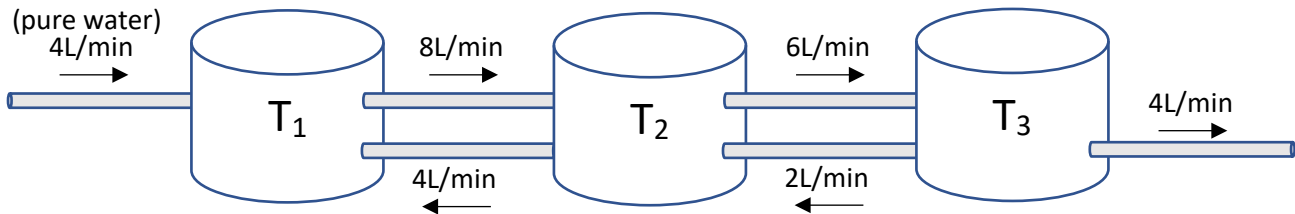
$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = - \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-3t}$$

(iii) (3 marks) Sketch the phase portrait of the above SDE.

Sketch the phase portrait in the box



(b) (4 marks) Set up a 3×3 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ of the following mixing tank problem with variables y_1 , y_2 and y_3 representing the amount of salt in the three tanks at time t respectively. The amount of water in each tank is 100 litre (L). (You are not required to solve the SDE.)



Working space

$$T_1: y_1' = \frac{4}{100}y_2 - \frac{8}{100}y_1$$

$$T_2: y_2' = \frac{8}{100}y_1 + \frac{2}{100}y_3 - \frac{10}{100}y_2$$

$$T_3: y_3' = \frac{6}{100}y_2 - \frac{6}{100}y_3$$

Write the underlying matrix in the box

$$\mathbf{A} = \begin{pmatrix} -0.08 & 0.04 & 0 \\ 0.08 & -0.10 & 0.02 \\ 0 & 0.06 & -0.06 \end{pmatrix}$$

Additional working space for Q19 (Indicate the parts clearly.)

Additional writing space (Indicate the question and part numbers clearly.)